
Animal Pose Estimation using Physics-Informed Neural Networks

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ABSTRACT Pose estimation — detecting the spatial configuration of a body’s joints from images or video — has seen tremendous progress in the human domain, but extending these techniques to animals remains significantly more challenging. Animals exhibit vastly different body plans across species, and most existing datasets consist of still images, making temporal modeling difficult. This paper presents a Physics-Informed Neural Network (PINN) approach for multi-species animal pose estimation over video data. A ResNet-18 backbone is used as the encoder, with a transposed convolution decoder producing 23-keypoint heatmaps decoded via soft-argmax. The model is trained on the Animal Kingdom dataset, which spans 850+ species across 7 taxonomic classes, using temporal triplets of consecutive frames. The total loss combines a visibility-masked data term with three physics-informed constraints: bone length rigidity, temporal smoothness, and joint angle plausibility. A two-stage inference pipeline integrating YOLOv8 detection improves keypoint localization for small animals. The model achieves 85.4% PCK on the validation set, exceeding the target of 70%, with validation loss converging from 0.027 to 0.0041 over 50 epochs. Results demonstrate that physics-informed constraints are a viable and lightweight addition to standard pose estimation pipelines, enforcing anatomical plausibility without significantly increasing model complexity.

INDEX TERMS animal pose estimation, physics-informed neural networks, keypoint detection, multi-species, temporal consistency, YOLOv8

I. INTRODUCTION

POSE estimation — detecting the spatial configuration of a body’s joints from images or video — has seen tremendous progress in the human domain, driven by large-scale datasets and deep convolutional architectures. However, extending these techniques to animals is significantly more challenging. Animals exhibit vastly different body plans: a fish, a frog, and a bird share almost no anatomical structure, yet a generalizable pose estimator must handle all of them.

The importance of animal pose estimation spans ecology (behavioral analysis), veterinary science (gait analysis), wildlife conservation (non-intrusive monitoring), and computer vision research (out-of-distribution generalization). Despite this, the field remains underexplored compared to human pose estimation, partly due to a lack of large, annotated, multi-species video datasets.

A. PROBLEM STATEMENT

Standard pose estimation models are designed for a single species (humans) under controlled conditions. Generalizing to 850+ animal species with drastically different body plans,

captured in the wild under varying lighting, scale, and occlusion, is an open problem. Furthermore, most animal pose datasets consist of still images, making temporal modeling — which is essential for video understanding — difficult to explore.

This project addresses the question: can a single neural network, augmented with physics-informed constraints, learn anatomically plausible pose predictions across hundreds of animal species from video data?

B. SCOPE OF THE PROJECT

The scope covers: dataset preprocessing and pipeline construction for the Animal Kingdom dataset [1], design of a PINN model architecture, formulation of physics-informed loss terms, GPU training on cloud infrastructure, and visual evaluation including integration with an off-the-shelf animal detector (YOLOv8 [6]). The project is intended as an academic exploration and does not aim to match state-of-the-art specialized systems.

II. OBJECTIVES

The following specific objectives were set at the outset of the project:

- Design and implement a PINN-based pose estimation model capable of handling multi-species animal video data.
- Formulate and validate physics-informed loss terms — bone length rigidity, temporal smoothness, and joint angle plausibility — that enforce physically realistic skeletal predictions across temporal triplets of frames.
- Train the model on the Animal Kingdom dataset (850+ species, 23 keypoints, 7 taxonomic classes) and achieve a PCK above 70% on the validation set.
- Build a complete, reproducible training pipeline with checkpoint saving, resume capability, and loss logging.
- Evaluate the model qualitatively by visualizing predicted skeletons on held-out animal frames across diverse species.
- Integrate YOLOv8 as a preprocessing detector to improve inference quality on frames where the animal occupies a small portion of the image.

III. LITERATURE REVIEW

A. HUMAN POSE ESTIMATION

The field of human pose estimation has been transformed by deep learning. DeepPose [9] introduced direct regression of joint coordinates via CNNs. Stacked Hourglass Networks [5] introduced multi-scale feature aggregation via heatmap prediction, becoming a dominant paradigm. HRNet [8] maintains high-resolution representations throughout, achieving state-of-the-art results on COCO benchmarks. These works established soft-argmax decoding and heatmap-based supervision as standard tools.

B. ANIMAL POSE ESTIMATION

Animal pose estimation has received comparatively less attention. LEAP [7] and DeepLabCut [4] addressed laboratory animal tracking under controlled conditions. APT-36K [10] extended this to 30 animal species. The Animal Kingdom dataset [1] is the most comprehensive, providing video clips and keypoint annotations across 850 species — making temporal modeling possible for the first time at this scale.

C. PHYSICS-INFORMED NEURAL NETWORKS

PINNs [2] were originally introduced for solving partial differential equations by embedding physical laws into the loss function. In pose estimation, physical priors take the form of structural constraints: bone lengths should remain constant (rigidity), joint positions should change smoothly over time (temporal consistency), and joint angles should fall within anatomically plausible ranges. These constraints have been explored in human motion capture but are relatively novel in animal pose estimation.

D. GAP AND MOTIVATION

No prior work combines PINN-style physical constraints with multi-species animal pose estimation over video. Most animal pose work focuses on still images or a single species. This project addresses this gap by building a temporally-aware, physics-constrained model for the full Animal Kingdom benchmark.

IV. METHODOLOGY

A. DATASET

The Animal Kingdom dataset [1] was used, specifically its pose estimation subset. Annotations are in MPII JSON format with 23 keypoints per frame. Frames are organized into video clips. A temporal triplet (t_0, t_1, t_2) is constructed from three consecutive frames of the same clip, forming the input unit for training. After filtering clips with fewer than 3 frames, 69,121 valid triplets were obtained from 83,187 total annotations across 7,157 clips.

B. MODEL ARCHITECTURE

The proposed model, *PoseCNN*, uses a ResNet-18 backbone [3] pretrained on ImageNet as the encoder, providing strong low-level feature extraction. The decoder consists of four transposed convolution layers that upsample the feature map back to the input resolution. The output is a heatmap of shape $(B, 23, H, W)$. Soft-argmax decoding converts heatmaps to continuous 2D keypoint coordinates, enabling end-to-end gradient flow. Total parameters: 13.9M.

C. LOSS FUNCTION

The total loss is a weighted combination of four terms:

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda_b \mathcal{L}_{\text{bone}} + \lambda_s \mathcal{L}_{\text{smooth}} + \lambda_a \mathcal{L}_{\text{angle}} \quad (1)$$

where $\lambda_b = 0.5$, $\lambda_s = 0.1$, and $\lambda_a = 0.05$.

- **Data Loss $\mathcal{L}_{\text{data}}$:** Visibility-masked MSE between predicted and ground-truth keypoint coordinates. Invisible keypoints (visibility = 0) are excluded from the loss.
- **Bone Loss $\mathcal{L}_{\text{bone}}$:** Mean squared error between predicted bone lengths and reference bone lengths estimated from the training set. Enforces anatomical rigidity across frames.
- **Smooth Loss $\mathcal{L}_{\text{smooth}}$:** L_2 penalty on the temporal difference of keypoint positions between consecutive frames in the triplet. Penalizes sudden jumps.
- **Angle Loss $\mathcal{L}_{\text{angle}}$:** Penalizes joint angles that fall outside anatomically plausible ranges estimated from training data.

D. TRAINING SETUP

The model was trained on a Kaggle Tesla T4 GPU (15 GB VRAM) using the Adam optimizer with an initial learning rate of 10^{-3} , cosine annealing after a 5-epoch warm-up. Batch size was 32, with `num_workers=4` and `pin_memory=True`. Training ran for 50 epochs with

5,000 triplets sampled per epoch. During the warm-up phase, the encoder was frozen (head-only training) to stabilize early optimization. Checkpoints were saved periodically and at the best validation loss, with resume functionality to handle Kaggle session drops.

E. INFERENCE PIPELINE

At inference time, YOLOv8n [6] detects the largest animal in the frame and returns a bounding box with 10% padding. The crop is resized to 256×256 and fed to PoseCNN. Predicted keypoints are then mapped back to the original image coordinate space for visualization.

V. EXPLORATION AND WORK DONE

A. DATASET SELECTION JOURNEY

The project originally targeted a COCO-format wildlife dataset with 11 species, 648 images, and 17 keypoints. After building an initial pipeline, a critical incompatibility was discovered: the dataset consisted of unordered still images with no temporal structure. Since the PINN loss requires consecutive frame triplets to compute smoothness and rigidity, this dataset was fundamentally incompatible with the architecture. This necessitated sourcing a new dataset entirely.

B. INITIAL PIPELINE — TRI-MOUSE

The pipeline was initially designed and tested on the TRI-MOUSE dataset (DLC CSV format, 12 keypoints, 3 mice tracked simultaneously). While this validated the triplet construction and temporal loss terms, the domain was too narrow (single species, lab setting) and the keypoint format required significant adaptation for the Animal Kingdom annotations.

C. INFRASTRUCTURE CHALLENGES

Significant time was spent navigating GPU access. The college GPU server (IIT BHU) was physically accessible from the lab, but the hostel network was on a different subnet, making SSH connections time out consistently. Multiple approaches were tried — direct SSH, SSH tunneling with port forwarding, Jupyter via browser — before settling on Kaggle as the primary training platform. A key lesson: always verify network connectivity before designing an infrastructure plan.

D. WINDOWS PATH BUG

The `dataset.py` file contained hardcoded Windows paths (`C:\Users\BHAVI\Downloads\...`) that caused immediate failures on Linux. These were patched programmatically via Python string replacement across multiple sessions — a tedious but instructive process that highlighted the importance of using relative paths or configuration files in research code.

E. KAGGLE SESSION MANAGEMENT

Kaggle imposes a 40-minute idle timeout and a 12-hour session limit. Training 50 epochs on 69,121 triplets initially

took 30+ minutes per epoch due to `num_workers=0` and `pin_memory=False`. This was resolved by setting `num_workers=4`, `pin_memory=True`, and limiting to 5,000 triplets per epoch — reducing epoch time to approximately 70 seconds. Resume functionality was added to `train.py` to handle session drops, and checkpoints were saved to Kaggle dataset storage for persistence across sessions.

F. SUCCESSFUL TRAINING

After resolving all infrastructure and path issues, the model trained successfully for 50 epochs on Kaggle. PCK rose from approximately 18% at epoch 1 to 85.4% at epoch 49. Loss convergence was clean across all components, with bone loss showing the strongest regularization effect after the warm-up phase ended.

VI. RESULTS AND ANALYSIS

A. QUANTITATIVE RESULTS

The model achieved 85.4% PCK on the validation set, exceeding the target of 70%. Validation loss converged from 0.027 at epoch 1 to 0.0041 at epoch 49. The bone loss component showed the strongest regularization effect among all physics-informed terms.

TABLE 1. Quantitative Performance Summary

Metric	Value	Notes	Status
Best PCK	85.4%	Epoch 49/50	✓ Achieved
Val Loss	0.0041	Converged	✓ Achieved
Total Triplets	69,121	5,000 used/epoch	✓ Achieved
Species Covered	850+	7 animal classes	✓ Achieved

B. QUALITATIVE RESULTS

Visual evaluation across diverse species revealed several patterns. Birds (pigeons, sparrows, woodpeckers) showed the strongest skeleton predictions, likely because they are well-represented in the training data. The model produced anatomically plausible skeletons for mammals (lions, deer) when detected correctly by YOLO. Performance was weaker for small or heavily occluded animals (insects, deep-sea creatures), where YOLO either failed to detect or produced large bounding boxes that diluted the crop.

C. EFFECT OF YOLOV8 INTEGRATION

Without detection-based cropping, keypoints for small animals were clustered near the image center. With YOLO cropping, the model operated on zoomed crops, producing significantly more spread-out and anatomically distributed keypoints. For cases where YOLO failed to detect an animal, the full image was used as fallback, resulting in degraded output.

VII. DISCUSSION

What Worked

The PINN loss formulation successfully enforced temporal consistency — predictions across the three frames of a triplet were smooth and physically plausible. The warm-up training strategy (head-only for 5 epochs) stabilized early training and prevented the encoder from drifting before the head was calibrated. The two-stage YOLO + PoseCNN pipeline significantly improved visual results without requiring any retraining of the pose model. Kaggle proved an effective training platform once epoch time was optimized, providing free GPU access with sufficient VRAM.

What Did Not Work

The initial COCO-format wildlife dataset was incompatible with the temporal architecture — a costly mismatch discovered late in the process. SSH-based access to the college GPU server from the hostel network was never successfully established due to subnet isolation. Training on all 69,121 triplets per epoch was impractical under Kaggle session constraints, necessitating subsampling to 5,000. YOLOv8 failed to detect exotic or small animals (e.g., insects, fish in turbid water), causing fallback to full-image mode with poor results.

Limitations

The model was evaluated only on validation data from the same distribution as training. Cross-dataset or held-out species evaluation was not performed. The 5,000 triplet/epoch subsampling means the model has seen only a fraction of the full dataset. YOLOv8 is not trained on all 850 Animal Kingdom species, creating a performance gap for rare taxa.

VIII. CONCLUSION

This project successfully designed, implemented, and trained a Physics-Informed Neural Network for multi-species animal pose estimation. Starting from a blank Python environment, a complete pipeline was built covering temporal triplet dataset construction, ResNet-18-based pose prediction, physics-motivated multi-term loss, and GPU training with checkpointing. The model achieved 85.4% PCK across 850+ species — a strong result that validates the PINN approach for this domain.

The project also demonstrated that infrastructure and data compatibility challenges are as significant as modeling choices in applied deep learning. Multiple failed approaches — wrong dataset format, subnet isolation, hardcoded paths, session timeouts — were encountered and resolved through iteration. Integrating YOLOv8 as a detection frontend improved inference quality without additional training cost.

All primary objectives were achieved. The broader implication is that physics-informed constraints are a viable

and relatively lightweight addition to standard pose estimation pipelines, adding anatomical plausibility without significantly increasing model complexity.

IX. FUTURE WORK

Future directions include:

- Train on the full 69,121 triplets for more epochs using multi-GPU DataParallel, which is already implemented but limited by Kaggle session time.
- Replace ResNet-18 with a stronger backbone (e.g., HRNet [8] or ViT-Pose) for improved feature representation.
- Fine-tune YOLOv8 [6] on Animal Kingdom species to improve detection accuracy for exotic taxa.
- Implement species-conditioned pose estimation, where a species embedding modulates the decoder to account for morphological differences.
- Evaluate on a held-out test set of unseen species to measure zero-shot generalization.
- Explore 3D pose lifting from 2D predictions using depth estimation or multi-view setups.

IMPORTANT LINKS

The following resources are publicly accessible and allow readers to reproduce or extend this work:

- Training Notebook (Kaggle): <https://www.kaggle.com/code/bhavika2008/notebook8917a292bb>
- Image Dataset (Kaggle): <https://www.kaggle.com/datasets/bhavika2008/ak-images>
- Code & Annotations Dataset (Kaggle): <https://www.kaggle.com/datasets/bhavika2008/ak-code-ann>
- Animal Kingdom Dataset (Original): <https://github.com/sutdcv/Animal-Kingdom>

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REFERENCES

- [1] K. Q. Ng et al., “Animal Kingdom: A Large and Diverse Dataset for Animal Behavior Understanding,” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, New Orleans, LA, USA, 2022.
- [2] M. Raissi, P. Perdikaris, and G. E. Karniadakis, “Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations,” *J. Comput. Phys.*, vol. 378, pp. 686–707, 2019.
- [3] K. He, X. Zhang, S. Ren, and J. Sun, “Deep Residual Learning for Image Recognition,” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, Las Vegas, NV, USA, 2016.
- [4] A. Mathis et al., “DeepLabCut: Markerless pose estimation of user-defined body parts with deep learning,” *Nature Neuroscience*, vol. 21, no. 9, pp. 1281–1289, 2018.
- [5] A. Newell, K. Yang, and J. Deng, “Stacked Hourglass Networks for Human Pose Estimation,” in *Proc. European Conf. Computer Vision (ECCV)*, Amsterdam, Netherlands, 2016.

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- [6] G. Jocher et al., “Ultralytics YOLOv8,” 2023. [Online]. Available: <https://github.com/ultralytics/ultralytics>
 - [7] T. D. Pereira et al., “Fast animal pose estimation using deep neural networks,” *Nature Methods*, vol. 16, no. 1, pp. 117–125, 2019.
 - [8] K. Sun, B. Xiao, D. Liu, and J. Wang, “Deep High-Resolution Representation Learning for Visual Recognition,” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, Long Beach, CA, USA, 2019.
 - [9] A. Toshev and C. Szegedy, “DeepPose: Human Pose Estimation via Deep Neural Networks,” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, Columbus, OH, USA, 2014.
 - [10] W. Ji et al., “APT-36K: A Large-scale Benchmark for Animal Pose Estimation and Tracking,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

Appendix

- Backbone: ResNet-18 (pretrained ImageNet)
- Keypoints: 23 per frame
- Input resolution: 256×256
- Batch size: 32
- Learning rate: 10^{-3} (cosine annealed to 10^{-5})
- Warm-up epochs: 5 (head-only)
- Loss weights: $\lambda_{\text{bone}} = 0.5$, $\lambda_{\text{smooth}} = 0.1$, $\lambda_{\text{angle}} = 0.05$
- Triplets per epoch: 5,000 (subsampling from 69,121)
- Total epochs: 50

- Platform: Kaggle Notebooks (free tier)
- GPU: NVIDIA Tesla T4 (2×, 15 GB VRAM each)
- Dataset storage: Kaggle Datasets (public)
- Dataset size: ~2 GB (images) + annotations

- Python 3.12, PyTorch 2.x, torchvision
- Ultralytics YOLOv8 (inference only)
- PIL, NumPy, Matplotlib
- Animal Kingdom dataset (Ng et al., 2022) [1]

BHAVIKA (B.Tech, Computer Science Engineering, IIT BHU Varanasi, Roll No: 24075016) conducted this project as an academic exploration in physics-informed deep learning for animal pose estimation. Her research interests include computer vision, pose estimation, and physics-informed machine learning. The project was submitted as an Exploratory Project Report to the Department of Computer Science and Engineering, Indian Institute of Technology (BHU) Varanasi, in April 2026.